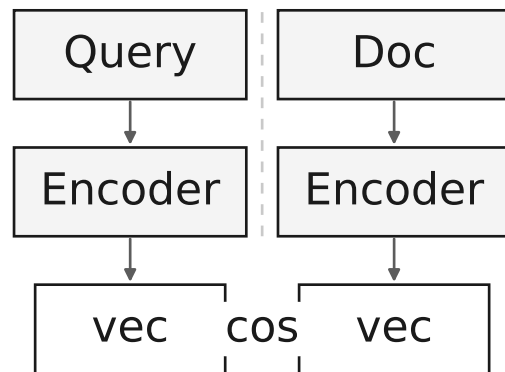


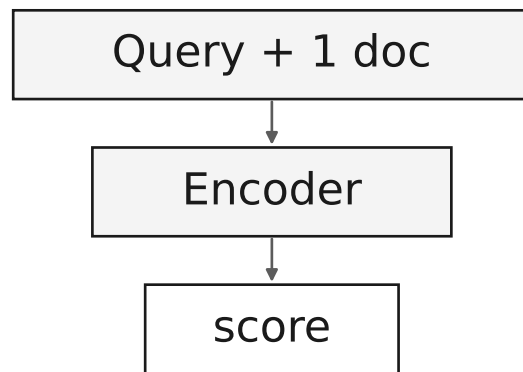
Bi-encoder

embedding pipeline



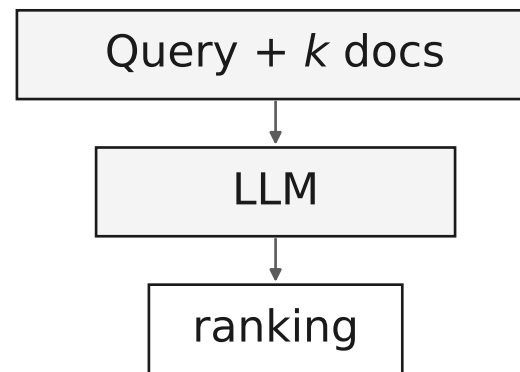
Cross-encoder

pairwise reranker



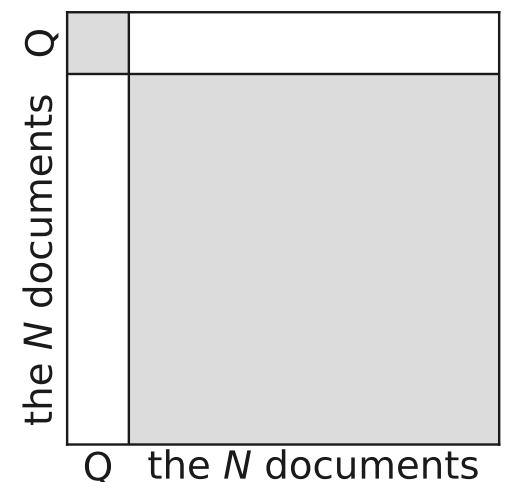
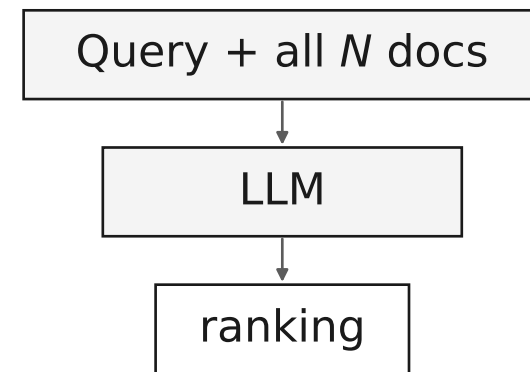
LLM listwise

reranker over top- k

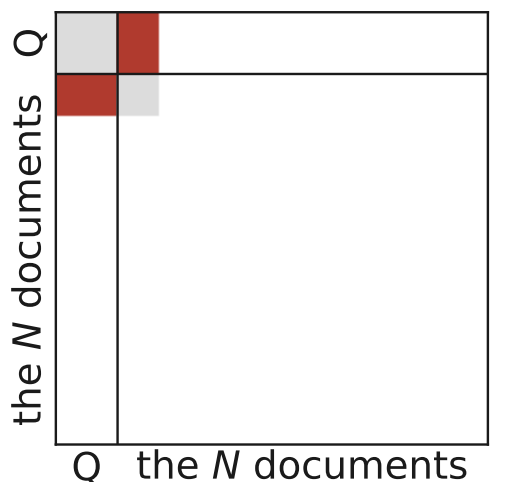


LLM corpus-in-context

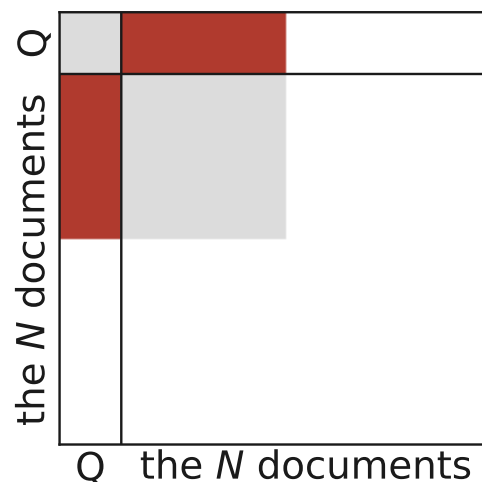
generative retrieval



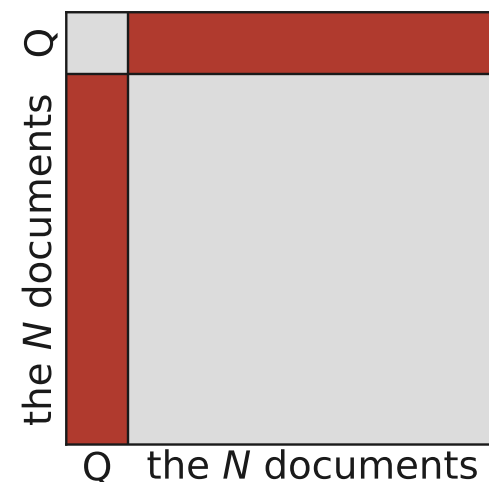
never jointly
 N vector comparisons
per query



one document at a time
 k pairwise passes
per query



k documents at once
one pass over k docs
per query



the whole corpus at once
one pass over N docs
per query

self-attention query-document no attention